

**POST AND TELECOMMUNICATIONS INSTITUTE OF TECHNOLOGY
FACULTY OF INFORMATION TECHNOLOGY 1**



FINAL REPORT BUSINESS INTELLIGENCE
Project: BI (Business Intelligence) analyzes e-commerce data

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1. Problem Introduction

1.1 Background

The rapid growth of e-commerce over the past decade has generated vast amounts of transactional data. Retailers operating online platforms now record every purchase, return, customer interaction, and product movement — creating data assets that, if properly analyzed, can drive significant competitive advantage. However, raw transactional data in isolation offers little strategic value. It must be structured, aggregated, and visualized before it can inform meaningful business decisions.

This project addresses that challenge by designing and implementing a Business Intelligence (BI) system for an e-commerce context. Rather than simply storing data, the goal is to transform raw sales records into actionable intelligence: understanding which products drive the most revenue, how sales fluctuate across time, which customer segments are most valuable, and where returns are concentrated.

1.2 Objectives

The primary objectives of this project are:

- To design a dimensional data model (Star Schema) suitable for OLAP-style analysis of e-commerce transactions
- To build an automated ETL pipeline that extracts, cleans, and loads raw data into a structured data warehouse
- To develop an interactive Power BI dashboard covering three analytical perspectives: product performance, revenue trends, and customer behavior
- To derive meaningful business insights from the data that could realistically support retail decision-making

1.3 Scope

The analysis covers transaction data spanning two full fiscal years (December 2009 to December 2011) from a UK-based online retailer serving customers across 40+ countries. The scope includes order-level analysis, product-level performance, time-series revenue trends, geographic distribution, and return rate monitoring. Machine learning and predictive modeling are outside the scope of this project.

2. Dataset Description

2.1 Source

The dataset used in this project is the **Online Retail II** dataset, published by the UCI Machine Learning Repository and originally contributed by Dr. Daqing Chen of the London South Bank University School of Engineering. It is publicly available at:

<https://archive.ics.uci.edu/dataset/502/online+retail+ii>

This dataset is widely used in academic research on retail analytics, customer segmentation, and demand forecasting, making it well-suited for a BI study.

2.2 Raw Data Profile

Attribute	Value
Format	Microsoft Excel (.xlsx), 2 sheets
Sheet 1	Year 2009–2010
Sheet 2	Year 2010–2011
Total raw records	1,067,371 rows
Number of columns	8
Time period	1 December 2009 – 9 December 2011
Geographic coverage	43 countries

2.3 Column Descriptions

Column	Renamed To	Data Type	Description
Invoice	InvoiceNo	String	Invoice number. Prefix 'C' indicates a cancellation/return
StockCode	StockCode	String	Unique product identifier
Description	Description	String	Product name
Quantity	Quantity	Integer	Number of units per transaction line
InvoiceDate	InvoiceDate	DateTime	Date and time of the transaction
Price	UnitPrice	Float	Unit price in GBP (£)

Column	Renamed To	Data Type	Description
Customer ID	CustomerID	String	Unique customer identifier (nullable)
Country	Country	String	Country of the customer

2.4. Data Quality Issues

Upon initial exploration, several data quality issues were identified that required resolution during the ETL phase:

Missing values:

CustomerID: 243,007 null values (22.8% of rows) — these represent guest checkouts without registered accounts

Description: 4,382 null values (0.4%)

Logical anomalies:

Negative Quantity values combined with 'C'-prefixed invoice numbers — these are legitimate return/cancellation records

UnitPrice values of 0.0 on non-return rows — likely internal adjustments or test entries, removed during cleaning

Non-standard country name encodings: "EIRE" for Ireland, "RSA" for South Africa, "USA" for United States

Volume context: After cleaning, 1,061,164 records (99.4% of raw data) were retained for analysis — indicating high overall data quality.

3. Data Warehouse Design

3.1 Design Philosophy

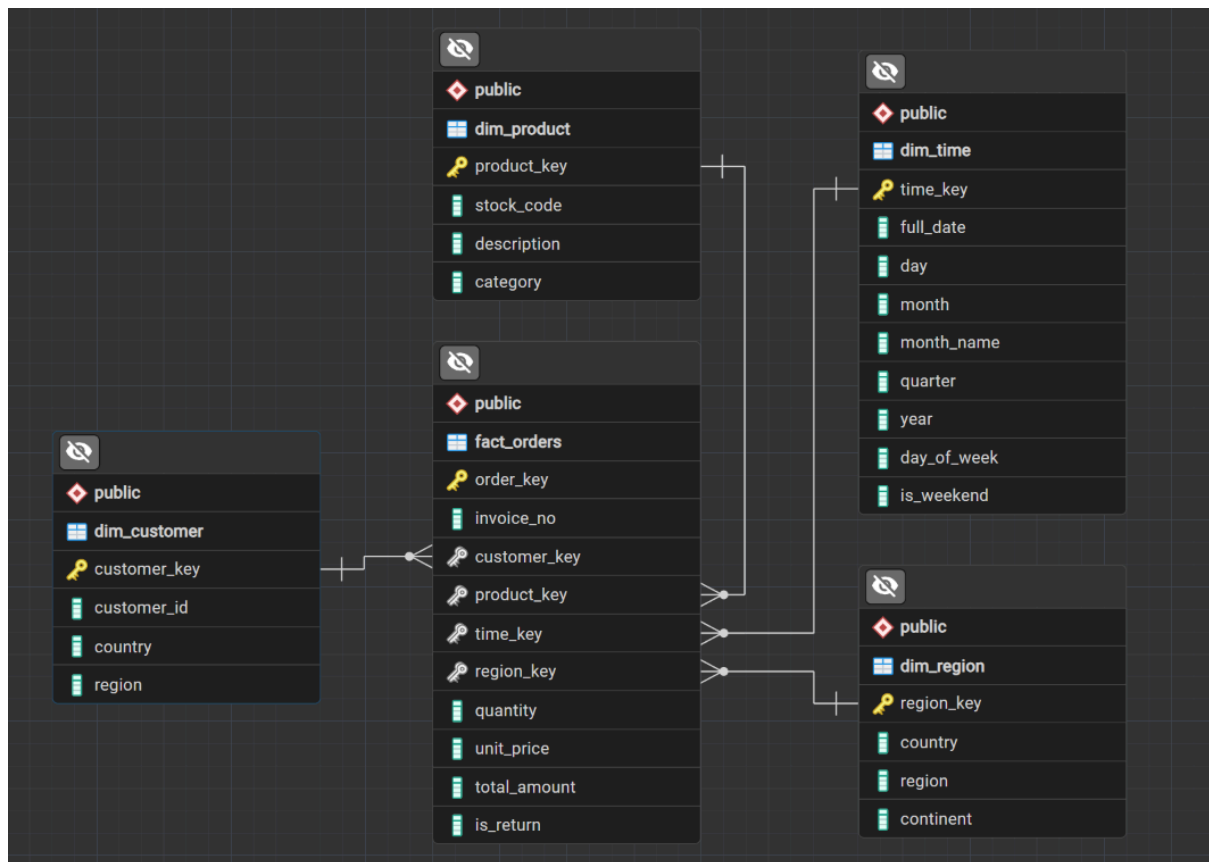
The data warehouse follows the **Star Schema** paradigm, as proposed by Ralph Kimball in his dimensional modeling methodology. This approach was selected over the normalized (3NF) alternative for several reasons:

Query performance: Denormalized dimension tables reduce the number of joins required for analytical queries

Comprehensibility: Business users and BI tools such as Power BI interact naturally with flat, descriptive dimension tables

Compatibility: Power BI's relationship engine is optimized for Star Schema structures

3.2 Schema Diagram



3.3 Fact Table: fact_orders

Each row in **fact_orders** represents one line item on an invoice — a single product within a single transaction. This grain was chosen because it preserves the maximum analytical flexibility: it supports both order-level and product-level aggregation.

Column	Type	Description
order_key	SERIAL PK	Auto-generated surrogate key
invoice_no	VARCHAR(20)	Original invoice identifier
customer_key	INT FK	Reference to dim_customer
product_key	INT FK	Reference to dim_product
time_key	INT FK	Reference to dim_time
region_key	INT FK	Reference to dim_region

Column	Type	Description
quantity	INT	Units sold (or returned if negative)
unit_price	NUMERIC(10,2)	Price per unit in GBP
total_amount	NUMERIC(12,2)	Computed: quantity × unit_price
is_return	BOOLEAN	True if invoice starts with 'C'

3.4 Dimension Tables

dim_customer records unique combinations of CustomerID and Country. Guest transactions (null CustomerID) are assigned the identifier "GUEST". The region column is derived from the country field.

dim_product stores one row per unique StockCode, with the most frequently occurring description selected as the canonical product name — resolving minor spelling variations in the source data.

dim_time was intentionally generated as a **continuous date range** from the minimum to maximum transaction date (1 December 2009 to 9 December 2011), producing 739 consecutive calendar days. This is a requirement for Power BI's time intelligence functions, which cannot operate correctly on date tables with gaps.

dim_region maps each country to a geographic region and continent, enabling hierarchical geographic drill-down in dashboards.

3.5 Analytical Views

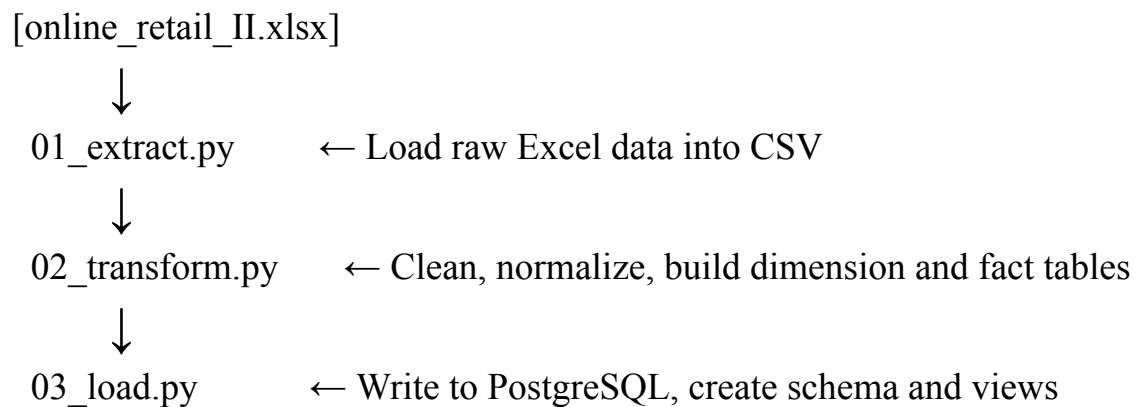
Five PostgreSQL views were created to support direct SQL querying and potential future integration with other reporting tools:

View	Purpose
vw_revenue_by_month	Monthly revenue, order count, and AOV aggregation
vw_top_products	Product performance ranked by revenue and quantity
vw_revenue_by_region	Geographic sales breakdown
vw_customer_behavior	Per-customer frequency and monetary metrics
vw_return_analysis	Monthly return rate tracking

4. ETL Process

4.1 Architecture Overview

The ETL pipeline was implemented entirely in Python using the Pandas data manipulation library and SQLAlchemy for database connectivity. The pipeline is organized into three sequential scripts:



4.2 Extract

The extraction script (`01_extract.py`) handles the multi-sheet Excel format of the source file. Both sheets are read simultaneously using `pd.read_excel(sheet_name=None)` and concatenated into a single DataFrame, producing 1,067,371 raw rows. All columns are initially read as strings to prevent premature type inference errors, particularly for CustomerID which pandas would otherwise parse as float (e.g., 12345.0).

The extracted data is persisted to `data/processed/retail_raw.csv` as an intermediate checkpoint, allowing the transform step to be re-run independently without re-reading the large Excel file.

4.3 Transform

The transformation step (`02_transform.py`) performs the following operations in sequence:

Column normalization: The source file uses different column names from the dataset's documentation (Invoice instead of InvoiceNo, Price instead of UnitPrice, Customer ID instead of CustomerID). A mapping dictionary renames these consistently.

Type casting: Quantity and UnitPrice are cast to numeric, with coercion of non-parseable values to NaN. InvoiceDate is parsed as datetime.

Return flagging: Rows where InvoiceNo begins with 'C' are flagged as `is_return = True`. This logic correctly identifies 23,564 cancellation records.

Filtering: Rows with `UnitPrice <= 0` on non-return records are removed (6,207 rows), as these represent internal adjustments rather than genuine transactions.

Country normalization: Non-standard country name variants are mapped to internationally recognized names:

EIRE → Ireland

RSA → South Africa

USA → United States

Channel Islands → United Kingdom

Dimension table construction: Each dimension table is built by extracting the relevant unique combinations of attributes, assigning integer surrogate keys, and enriching with derived attributes (e.g., region mapped from country, time attributes derived from InvoiceDate).

Fact table construction: Foreign key lookup dictionaries are built from each dimension table, and the fact table is assembled by mapping original business keys to surrogate keys. Rows with unmappable product or time keys are dropped.

4.4 Load

The load script (`03_load.py`) connects to PostgreSQL using credentials stored in a `.env` file. The schema creation SQL is executed first (`01_create_schema.sql`), which drops and recreates all tables — making the pipeline idempotent and safe to re-run.

Dimension tables are loaded before the fact table to satisfy foreign key constraints. The fact table (1,061,164 rows) is loaded in chunks of 5,000 rows using SQLAlchemy's `to_sql` with `method="multi"` for bulk insert efficiency.

4.5 Pipeline Results

Table	Row Count	Notes
<code>dim_region</code>	41	43 countries after merging normalized duplicates
<code>dim_customer</code>	5,967	Unique customer × country combinations
<code>dim_product</code>	4,932	Unique stock codes

Table	Row Count	Notes
dim_time	739	Continuous: 2009-12-01 to 2011-12-09
fact_orders	1,061,164	99.4% retention from raw data

5. Dashboard Development

5.1 Tool Selection

Power BI Desktop was selected as the visualization platform. Its native PostgreSQL connector enabled direct import of the data warehouse tables without requiring an intermediate file export. The Import connectivity mode was used, loading all data into Power BI's in-memory engine for maximum query performance.

5.2 Data Model in Power BI

After importing the five tables, relationships were established in Power BI's Model view:

From (Fact)	To (Dimension)	Cardinality	Direction
fact_orders.customer_key	dim_customer.customer_key	Many-to-One	Single
fact_orders.product_key	dim_product.product_key	Many-to-One	Single
fact_orders.time_key	dim_time.time_key	Many-to-One	Single
fact_orders.region_key	dim_region.region_key	Many-to-One	Single

dim_time was marked as the official Date Table using full_date as the date column, enabling Power BI's built-in time intelligence functions.

5.3 DAX Measures

Six core measures were defined in DAX to support dashboard calculations:

Total Revenue =

CALCULATE(SUM(fact_orders[total_amount]), fact_orders[is_return] = FALSE)

Total Orders =

CALCULATE(DISTINCTCOUNT(fact_orders[invoice_no]),

fact_orders[is_return] = FALSE)

Total Quantity =

CALCULATE(SUM(fact_orders[quantity]), fact_orders[is_return] = FALSE)

AOV = DIVIDE([Total Revenue], [Total Orders])

Return Rate =

DIVIDE(

CALCULATE(COUNTROWS(fact_orders), fact_orders[is_return] = TRUE),
COUNTROWS(fact_orders)

) * 100

Cumulative Revenue =

CALCULATE(

[Total Revenue],

FILTER(ALL(dim_time[full_date]), dim_time[full_date] <=

MAX(dim_time[full_date]))

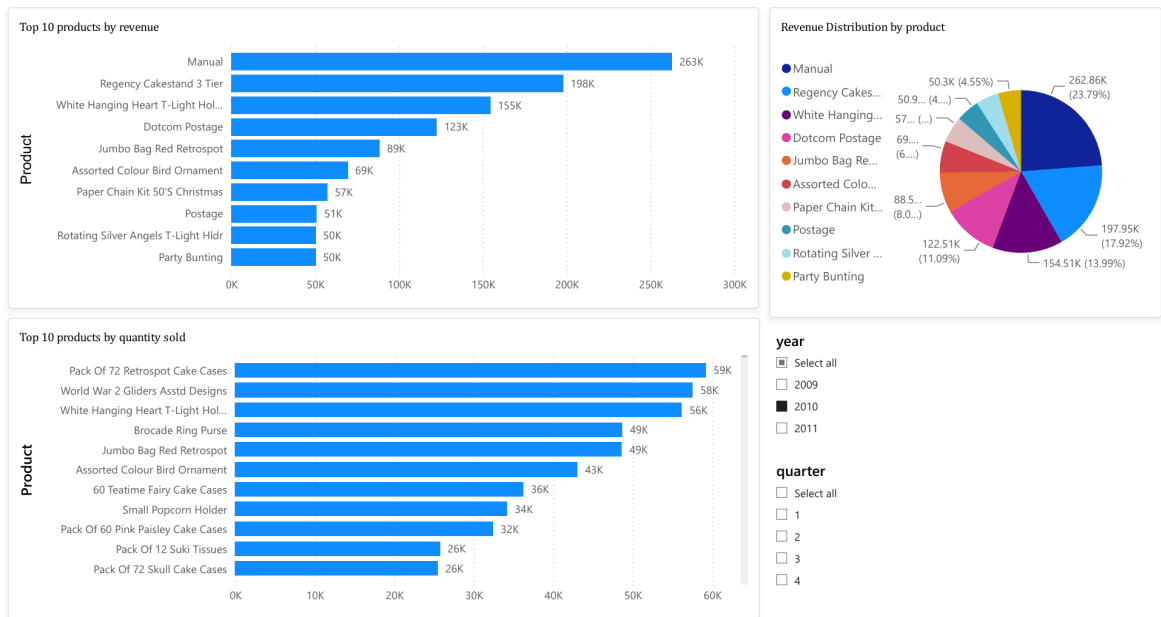
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5.4 Dashboard Pages

The dashboard is organized into three pages, each targeting a distinct analytical question.

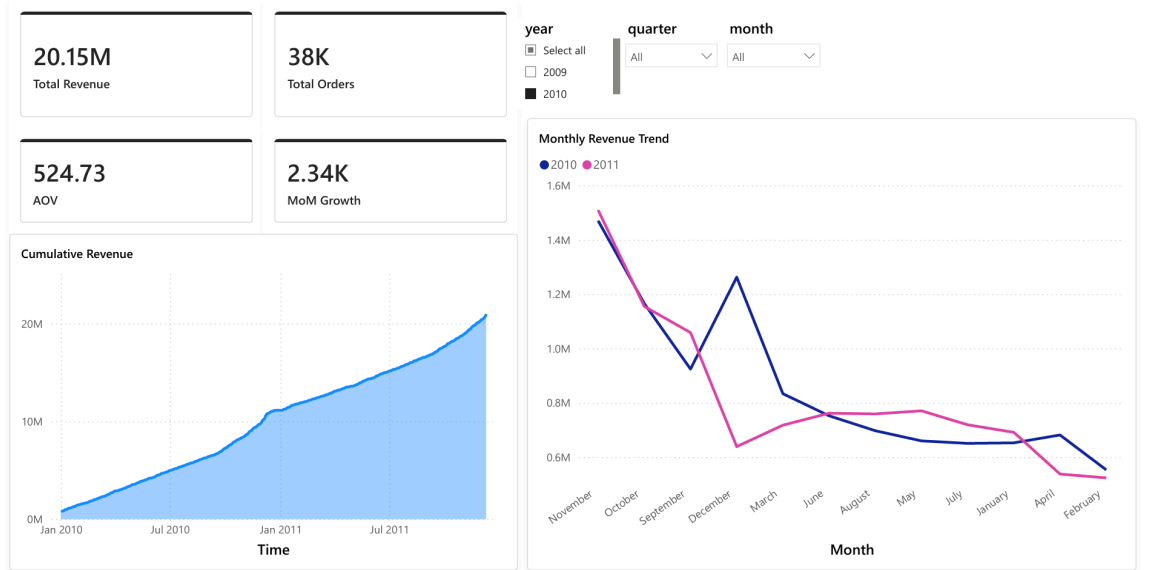
Page 1: Top Products

Visual	Type	Fields
Top 10 Products by Revenue	Horizontal Bar Chart	Description vs. Total Revenue (Top N = 10)
Top 10 Products by Quantity	Horizontal Bar Chart	Description vs. Total Quantity (Top N = 10)
Revenue Distribution by Product	Pie chart	Description, Total Revenue
Year Filter	Slicer (List)	dim_time[year]
Quarter Filter	Slicer (List)	dim_time[quarter]



Page 2: Monthly Revenue

Visual	Type	Fields
Monthly Revenue Trend	Line Chart	month_name vs. Total Revenue, by year
Total Revenue	KPI Card	Total Revenue measure
Total Orders	KPI Card	Total Orders measure
Average Order Value	KPI Card	AOV measure
Cumulative Revenue	Area Chart	full_date vs. Cumulative Revenue



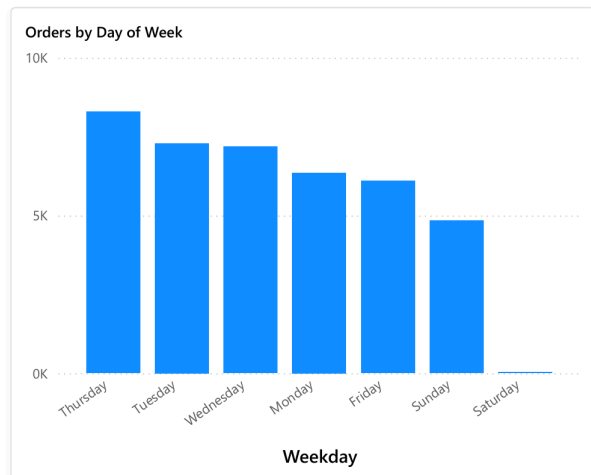
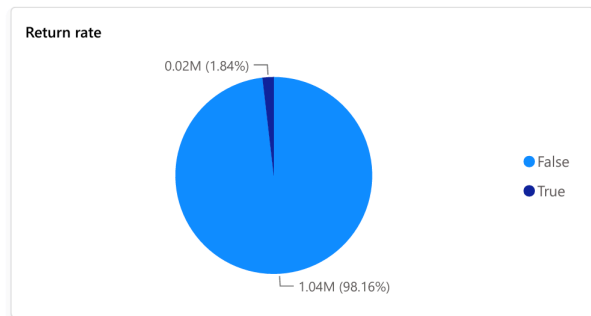
Page 3: Customer Behavior

Visual	Type	Fields
Revenue by Country	Map	country (location), Total Revenue
Return Rate	Pie Chart	is_return (legend), Order Count
Top Customers	Table	customer_id, country, Total Orders, Total Revenue
Orders by Day of Week	Column Chart	day_of_week vs. Total Orders



Top customers

customer_id	country	Total Orders	Total Revenue
GUEST	United Kingdom	2994	3,147,830.26
18102	United Kingdom	145	608,821.65
14646	Netherlands	151	528,602.52
14156	Ireland	156	313,946.37
14911	Ireland	398	295,972.63
17450	United Kingdom	51	246,973.09
13694	United Kingdom	143	196,482.81
17511	United Kingdom	60	175,603.55
16446	United Kingdom	2	168,472.50
16694	United Kingdom	55	147,142.77
Total		40077	20,972,594.55



6. Insight Analysis

The following analysis is drawn directly from the data warehouse. All figures are exact values computed from the full dataset (December 2009 – December 2011) via PostgreSQL queries, unless a specific year filter is noted.

Overall KPI summary:

Metric	Value
Total Revenue (excl. returns)	£20,972,594.55
Total Orders (distinct invoices)	40,077
Average Order Value	£523.30
Return Rate	1.84% (19,543 of 1,061,164 lines)

6.1 Product Performance

Top 10 Products by Revenue (all years, returns excluded):

Rank	Product	Revenue
1	Manual	£206,245.48

Rank	Product	Revenue
2	Regency Cakestand 3 Tier	£174,398.30
3	White Hanging Heart T-Light Holder	£100,099.60
4	Rabbit Night Light	£99,808.80
5	Dotcom Postage	£95,601.60
6	Paper Craft , Little Birdie	£90,475.93
7	Picnic Basket Wicker 60 Pieces	£88,929.60
8	Jumbo Bag Red Retrospot	£87,897.50
9	Assorted Colour Bird Ornament	£64,785.63
10	Rotating Silver Angels T-Light Hldr	£61,985.73

Top 10 Products by Quantity Sold (all years):

Rank	Product	Units Sold
1	World War 2 Gliders Asstd Designs	110,138
2	Jumbo Bag Red Retrospot	98,768
3	White Hanging Heart T-Light Holder	96,744
4	Pack Of 72 Retrospot Cake Cases	96,560
5	Small Popcorn Holder	89,898
6	Assorted Colour Bird Ornament	81,809
7	Paper Craft , Little Birdie	80,995
8	Medium Ceramic Top Storage Jar	78,033
9	Brocade Ring Purse	71,430
10	Pack Of 60 Pink Paisley Cake Cases	56,794

A comparison of the two rankings reveals an important strategic distinction. "World War 2 Gliders Asstd Designs" tops the volume chart with 110,138 units sold, yet does not appear in the top-10 by revenue — a clear indicator of its low unit price positioning. Conversely, "Manual" leads the revenue chart while not appearing in the volume rankings, suggesting it is a high-ticket, low-frequency item. "White Hanging Heart T-Light Holder" and "Assorted Colour Bird Ornament" appear prominently in both lists, representing the most balanced performers in terms of both volume and margin.

This divergence has direct implications for inventory and pricing strategy: high-revenue items require supply security and margin protection, while high-volume items demand efficient fulfillment capacity and reorder discipline.

6.2 Revenue Trends

Monthly revenue figures across all years reveal a consistent and pronounced seasonal pattern:

Month	2009 Revenue	2010 Revenue	2011 Revenue
January	—	£652,708.50	£691,364.56
February	—	£553,339.73	£523,631.89
March	—	£833,570.13	£717,639.36
April	—	£681,528.99	£537,808.62
May	—	£659,858.86	£770,536.02
June	—	£752,270.14	£761,739.90
July	—	£650,712.94	£719,221.19
August	—	£697,274.91	£759,138.38
September	—	£924,333.01	£1,058,590.17
October	—	£1,165,483.91	£1,154,979.30
November	—	£1,470,272.48	£1,509,496.33
December	£825,685.76	£1,262,598.79	£638,810.68*

**December 2011 truncated at 9 December.*

November is the revenue peak in both 2010 and 2011, reaching £1,470,272 and £1,509,496 respectively — approximately 2.7× the February trough in each year. This extreme seasonality is consistent and structural, not anomalous. The total cumulative revenue across the observation period is **£20,972,594.55**, with the final quarter (Q4) contributing a disproportionately large share in both years.

Year-over-year growth is visible in the Q3–Q4 period: September grew from £924,333 (2010) to £1,058,590 (2011), a 14.5% increase. November grew from £1,470,272 to £1,509,496 (+2.7%). These figures suggest the business was on a moderate upward trajectory heading into late 2011.

6.3 Geographic Distribution

The United Kingdom overwhelmingly dominates revenue, accounting for **£17,915,974.52 (85.4%)** of total revenue from **36,590 orders** (91.3% of all orders). The top 15 countries by revenue are:

Rank	Country	Revenue	Orders
1	United Kingdom	£17,915,974.52	36,590
2	Ireland	£664,431.78	626
3	Netherlands	£554,232.34	228
4	Germany	£432,562.71	793
5	France	£356,944.60	622
6	Australia	£169,968.11	95
7	Spain	£109,178.53	154
8	Switzerland	£101,011.29	93
9	Sweden	£91,903.72	105
10	Denmark	£69,862.19	43
11	Belgium	£65,753.42	149
12	Norway	£60,109.62	45
13	Portugal	£58,016.65	95
14	Japan	£47,138.39	33
15	Italy	£32,550.42	65

Ireland (£664,431) and the Netherlands (£554,232) are the top international markets, both exceeding half a million pounds. Notably, the Netherlands achieved this with only 228 orders — a very high average order value of £2,430 per invoice, strongly suggesting wholesale purchasing patterns. Denmark similarly shows only 43 orders yet £69,862 in revenue (£1,625 per order average).

6.4 Customer Behavior

Orders by day of week reveal a sharp weekday concentration:

Day	Orders	% of Total
Thursday	8,288	20.7%
Tuesday	7,291	18.2%
Wednesday	7,185	17.9%
Monday	6,351	15.8%
Friday	6,097	15.2%
Sunday	4,835	12.1%
Saturday	30	0.07%

Saturday accounts for just 30 orders — effectively zero activity. This extreme weekend drop-off is consistent with a predominantly B2B wholesale customer base: buyers place orders during business hours on weekdays. Saturday represents less than 0.1% of all orders, compared to Thursday's 20.7%. This insight directly informs staffing and system maintenance scheduling decisions.

Return rate: 19,543 transaction lines (1.84% of all lines) are flagged as returns. The remaining 1,041,621 lines (98.16%) are regular sales. This return rate is notably low — industry benchmarks for general e-commerce returns typically range from 20–30%. The low figure reflects the giftware and home décor product mix, where purchase intent is strong, as well as the wholesale buyer profile.

Top customers by revenue:

Rank	Customer ID	Country	Orders	Revenue
1	GUEST	United Kingdom	2,994	£3,147,830.26
2	18102	United Kingdom	145	£608,821.65
3	14646	Netherlands	151	£528,602.52
4	14156	Ireland	156	£313,946.37
5	14911	Ireland	398	£295,972.63
6	17450	United Kingdom	51	£246,973.09
7	13694	United Kingdom	143	£196,482.81
8	17511	United Kingdom	60	£175,603.55

Rank	Customer ID	Country	Orders	Revenue
9	16446	United Kingdom	2	£168,472.50
10	16684	United Kingdom	55	£147,142.77

The guest segment (unregistered transactions) is the single largest revenue contributor at £3,147,830 across 2,994 orders — representing 15.0% of total revenue flowing through anonymous transactions. Among registered customers, Customer 18102 (UK) leads with £608,821 from 145 orders (average order value: £4,199). Customer 16446 (UK) presents the most striking case: just 2 orders totaling £168,472 — an average of £84,236 per invoice, indicating a single large bulk wholesale transaction.

The top 10 registered customers alone account for approximately £2.68M in revenue, highlighting the importance of key account retention in this business.

6.5 Key Takeaways

Finding	Evidence	Business Implication
Extreme Q4 seasonality	November £1.47–1.51M vs. February £0.52–0.55M	Begin inventory ramp-up in August; concentrate marketing spend in Oct–Nov
Saturday is operationally inactive	30 orders on Saturday vs. 8,288 on Thursday	Schedule system maintenance, updates, and staff rest on Saturdays
UK generates 85.4% of revenue	£17.9M of £20.97M total	Domestic operations are the core — but Netherlands (£2,430 avg order) shows wholesale export potential
Guest segment = 15% of total revenue	£3.15M from 2,994 anonymous orders	Account registration incentives could convert these buyers to trackable, loyalty-eligible customers
Return rate is very low (1.84%)	19,543 returns of 1,061,164 lines	Product quality and description accuracy are

Finding	Evidence	Business Implication
		adequate; no immediate corrective action needed
High-value wholesale accounts exist	Customer 16446: £168K from 2 orders; 18102: £609K from 145	Dedicated key account management and pricing agreements warranted for top-tier buyers
Revenue $\neq$ Volume leaders	"Manual" tops revenue; "World War 2 Gliders" tops volume	Separate procurement, pricing, and promotion strategies required for each product tier

6.6 Product Performance

The top products by revenue in 2010 reveal a clear hierarchy of high-value items. The leading product, **"Manual"**, generated £262,860 — accounting for 23.79% of the top-10 revenue pool on its own, an unusually high concentration for a single SKU. The next tier includes the **Regency Cakestand 3 Tier** (£197,950 / 17.92%) and the **White Hanging Heart T-Light Holder** (£154,510 / 13.99%). Rounding out the top five are **Dotcom Postage** (£122,510) and **Jumbo Bag Red Retrospot** (£88,500).

The top products by **quantity sold** tell a markedly different story. The highest-volume item, **Pack Of 72 Retrospot Cake Cases**, moved 59,000 units in 2010 — yet does not appear in the top-10 revenue chart, reflecting its low unit price. Similarly, **World War 2 Gliders Asstd Designs** (58,000 units) and **White Hanging Heart T-Light Holder** (56,000 units) are volume leaders. The White Hanging Heart T-Light Holder is the only item appearing prominently in both rankings, suggesting it achieves a favorable balance of volume and margin. This divergence between revenue leaders and volume leaders underscores the importance of examining both dimensions separately when making stocking, promotional, and pricing decisions. High-revenue items warrant protection of supply and margin; high-volume items require inventory depth and efficient fulfillment logistics.

6.7 Revenue Trends

The monthly revenue line chart reveals strong and consistent **Q4 seasonality** across both 2010 and 2011. The following figures illustrate the pattern for 2010:

Month	2010 Revenue	2011 Revenue
January	~£0.70M	~£0.70M
February	~£0.63M	~£0.60M
March	~£0.84M	~£0.78M
April	~£0.73M	~£0.60M
May	~£0.70M	~£0.79M
June	~£0.78M	~£0.78M
July	~£0.71M	~£0.72M
August	~£0.72M	~£0.80M
September	~£0.90M	~£0.98M
October	~£1.08M	~£1.10M
November	~£1.50M	~£1.52M
December	~£1.35M	~£0.70M*

**December 2011 is truncated as the dataset ends on 9 December 2011.*

November is the peak revenue month in both years, reaching approximately £1.50–1.52M — more than double the trough months of January–February. This consistent pattern across two independent years confirms that the seasonality is structural rather than coincidental. The cumulative revenue curve reaches a total of **£20.15M** by the end of the observation period, with the steepest gradient occurring between September and November each year.

The year-over-year trajectory shows modest growth: Q3 and Q4 revenues in 2011 slightly outpace their 2010 equivalents (e.g., September: £0.90M → £0.98M; October: £1.08M → £1.10M), suggesting the business was on a gradual upward trend.

6.8 Geographic Distribution

The filled map visual confirms that the **United Kingdom** is the dominant market by a substantial margin, which is expected for a UK-based retailer. Beyond the domestic market, the top international contributors include Ireland, the Netherlands, and other Western European nations.

The top customers by country reveal that the highest-value individual accounts outside the UK are concentrated in **Ireland** and the **Netherlands**. Customer 14646 (Netherlands) generated £528,602 across 151 orders, while customers 14156 and 14911 (both Ireland) contributed £313,946 and £295,972 respectively.

The map also shows smaller clusters of orders in the United States, Australia, South Africa, and Japan — representing markets where the retailer has some international reach but has not established dominant volume.

6.9 Customer Behavior

Order frequency by day of week shows a pronounced weekday bias:

Day	Approximate Orders
Thursday	~8,000 (highest)
Tuesday	~7,200
Wednesday	~7,000
Monday	~6,600
Friday	~6,400
Sunday	~4,300
Saturday	~50 (negligible)

Saturday is essentially inactive, with only ~50 orders recorded — less than 1% of Thursday volume. This pattern strongly suggests a **B2B wholesale customer base**: buyers are placing orders during standard business hours on weekdays. Saturday represents a near-complete operational shutdown from the customer side. This insight has direct implications for staffing, order processing scheduling, and customer communication timing.

Return rate analysis shows that 1,040,000 transaction lines (98.16%) are normal sales, with only approximately 20,000 lines (1.84%) flagged as returns. This is a notably low return rate for retail — typical e-commerce fashion return rates can exceed 30%. The low figure likely reflects the giftware and home décor product mix, where customers have strong prior intent before purchasing, as well as the predominance of wholesale buyers who inspect goods upon delivery rather than returning them post-sale.

Top customer analysis reveals significant revenue concentration. The top individual identified account — Customer 18102 (UK) — generated £608,821 from just 145 orders, implying an extremely high average order value of

approximately £4,199 per invoice. Customer 16446 (UK) presents an even more extreme case: only 2 orders, yet £168,472 in total revenue — suggesting a single large wholesale purchase. These accounts represent significant business relationships that would warrant dedicated account management.

The "GUEST" segment (unregistered customers) collectively placed 2,994 orders totaling £3,147,830 — the highest revenue contribution of any single entry. This highlights that a substantial portion of revenue flows through untracked transactions, which limits the retailer's ability to build loyalty programs or targeted marketing for this segment.

6.10 Key Takeaways

Finding	Specific Evidence	Business Implication
Strong Q4 seasonality	November peaks at £1.50–1.52M vs. February troughs of £0.60–0.63M	Begin inventory build-up in August; allocate marketing budget heavily toward Q4
Revenue concentration in top SKUs	"Manual" alone = 23.79% of top-10 revenue in 2010	Protect supply continuity for top-performing SKUs; avoid stockouts
Volume ≠ Revenue leaders	Pack Of 72 Cake Cases: 59K units but not in top-10 revenue	Separate volume strategy from margin strategy in procurement
Saturday is operationally dormant	~50 orders on Saturday vs. ~8,000 on Thursday	Schedule maintenance, system updates, and logistics rests for Saturday
Return rate is very low (1.84%)	1.04M normal vs. 0.02M returned	Current product quality and description accuracy are adequate
Guest segment is the largest single revenue contributor	£3.15M from guest checkouts	Incentivize account registration through loyalty discounts or order history access
High-value wholesale accounts exist	Customer 18102: £608K from 145 orders; Customer 16446: £168K from 2 orders	Implement key account management for top-tier buyers

7. Conclusion

7.1 Summary

This project successfully delivered a complete Business Intelligence system for e-commerce transaction analysis. Starting from a publicly available dataset of over one million raw records, the project produced a structured data warehouse built on a Star Schema in PostgreSQL, populated through a reproducible Python ETL pipeline, and visualized through a three-page Power BI dashboard.

The system supports analysis across three key dimensions: product performance, temporal revenue trends, and customer/geographic behavior. Each component was designed with practical usability in mind — the ETL pipeline is modular and re-runnable, the schema is optimized with indexes for analytical queries, and the dashboard provides interactive filtering through slicers.

Github source-code: <https://github.com/7gho/e-commerce-bi-project.git>

7.2 Limitations

Several limitations should be acknowledged:

Category data: All products were assigned to a single generic category (General) due to the absence of category information in the source dataset. A more granular product taxonomy would significantly enhance product-level analysis.

Customer demographics: The dataset contains no demographic information beyond country, limiting the depth of customer segmentation that can be performed.

Guest transactions: 22.8% of transactions have no associated CustomerID, meaning a significant share of revenue cannot be attributed to specific customer journeys.

Static dataset: The analysis is retrospective and covers a fixed historical window. The system does not currently support incremental data loading or real-time refresh.

7.3 Future Work

The following extensions would materially improve the system:

Product categorization: Apply rule-based keyword matching or clustering to assign products to meaningful categories based on their descriptions

RFM segmentation: Implement a full Recency-Frequency-Monetary segmentation model to classify customers into actionable tiers

(Champions, Loyal, At Risk, Lost)

Forecasting: Integrate time-series forecasting (e.g., ARIMA or Prophet) to project future revenue and inform inventory planning

Incremental ETL: Redesign the pipeline to support delta loading — appending only new records rather than full reloads

Data quality monitoring: Add automated data quality checks and logging to detect anomalies in source data before they affect the warehouse

8. References

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